

Dexter Fernandes

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Computer Vision Engineer

Computer vision engineer with 3.5 years shipping production machine learning models in Python and PyTorch. Owned the full ML lifecycle at safety-critical deployments in Austin, Texas and Oman: large-scale image dataset curation, model training, quantisation, and edge inference running 24/7 unattended on BP drilling rigs. Built a two-stage detector running on the edge and in the cloud that cut false positives by 33% on dual-camera setups mounted to construction vehicles.

WORK EXPERIENCE

GrowthStage • Remote Recruiter

05/2026 - Present

- Evaluated candidates' technical depth across perception, SLAM, and deployment using hands-on robotics and computer vision knowledge.

Career Break Self-directed

09/2024 - 04/2026

- [Enter relevant info here]

Esbaar • Muscat, Oman AI Engineer

01/2023 - 08/2023

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, covering data curation, training, deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 3 days to under 24 hours.
- Generated synthetic images and augmentation routines to cover edge cases in OpenCV and scikit-image to cover harsh industrial conditions, reducing false positives by 13%.
- Built benchmarking harnesses comparing YOLOv5 and EfficientDet on accuracy, latency and stability before and after optimisation.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.

LivNSense Technologies • Bengaluru, India Lead Software Engineer - AI

05/2020 - 12/2022

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO and Intel Movidius VPUs, raising throughput from 5 to 15 FPS and holding deterministic real-time inference on Intel edge hardware.
- Designed CNN classifiers in PyTorch to infer vehicle state from optical flow in production, supporting collision avoidance on moving vehicles.
- Deployed quantised vision models using FP16/INT8 optimisation, pruning and layer fusion, reducing model size by 50% while preserving production accuracy targets.
- Automated labelling by repurposing existing detection models and added dataset versioning using DVC, cutting manual labelling effort by 80%.

- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.

PROJECTS

DeepStream RTSP Pipeline - <https://github.com/Dexter-Fernandes/deepstream-rtsp-pipeline>

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDFI, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.

Modern Edge CV Benchmarking - <https://github.com/Dexter-Fernandes/Modern-Edge-CV-Benchmarking>

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 on a Clearpath Husky for the dissertation, fusing Velodyne VLP-16 LiDAR, RealSense depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences to compare mapping and localisation behaviour under a single evaluation protocol.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, validating sensor placement for coverage.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

SKILLS

- **Computer Vision & Deep Learning:** Anomaly detection, Convolutional Neural Networks, EfficientDet, Image classification, Keras, NvDCF, NvSort, Object detection, Object tracking, OpenCV, Optical Flow, PyTorch, RT-DETR, Scikit-image, Scikit-Learn, Segmentation, TensorFlow, Torchvision, YOLO
- **Edge Inference and Optimisation:** CUDA C++, custom TensorRT plugins, FP16/INT8 Quantisation, GStreamer, Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA DeepStream, NVIDIA Jetson AGX Xavier, ONNX, OpenVINO, Quantisation-Aware Training, TensorRT, VRAM profiling
- **Programming, Cloud & MLOps:** Apache Airflow, Apache Kafka, C++17, CI/CD, CMake, containerised microservices, Docker, DVC (Data Version Control), Git, gtest, Jupyter, Linux Bash, Microsoft Azure, MLflow, Optuna, PIL (Pillow), pytest, Python, REST APIs, SQL (PostgreSQL), Weights & Biases (wandb)
- **Production Systems:** Cost-per-inference, Health checks, Model-promotion gates, Observability

- **Annotation & Interpretability:** CVAT (Computer Vision Annotation Tool), GRAD-CAM, Synthetic data generation, TIDE (Toolbox for Identifying Detection Errors)
- **Edge AI & Video Pipelines:** Camera behaviour debugging, FFmpeg, GStreamer, ISP tuning, Multi-stream video analytics, RTSP video pipelines